Check what you know

You have learned

- how to plan a program
- what an algorithm is
- what happens when you run a program.



Activity

Here is a plan for the actions in a quiz program. They are not sorted into order.



Tell you if the answer is right

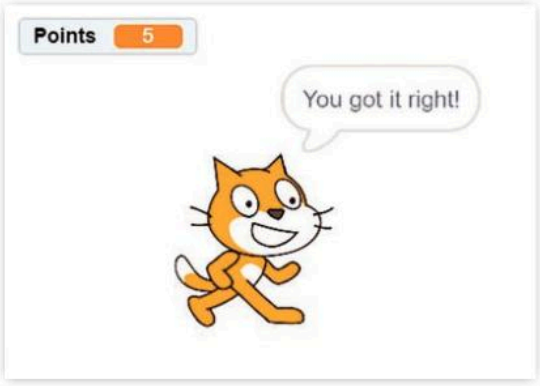
Show how many points you have got

Ask a question

Give you a point for a right answer

Get the answer to the question

Make an algorithm by putting the actions in the right order.



Test

A programmer made a program. Here are the three things she did.

Plan the program
Write the program
Run the program

1. Write the three actions in your book.
2. Tick the action that means carry out the commands.
3. Put a star next to the one that means make an algorithm.
4. Fill in the missing words in these sentences. Use the words from the box.

actions	algorithm	sequence	program

An _____ is a plan to make a _____.

It sets out all the _____ in the right _____.

Self-evaluation

- I answered test questions 1 and 2.
- I answered test questions 1–3.
- I did some of the activity.
- I answered all the test questions.
- I did all of the activity.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?

4 Programming: The Frog Hop Game

You will learn

- how a program is made of commands
- how to pick commands to make the program you want
- how to make a program that works
- how to make changes and corrections to a program.

In this unit you will make a computer program. You will make a new computer game called the Frog Hop Game. You will use Scratch.



Class activity

Here is a picture of the Scratch screen:



Learning outcomes: Say what a program will do by looking at its commands; Create a simple program and remove errors so it runs

The different parts of the screen are:

- the code blocks: the blocks are stored here ready to use
- the script area: this is where you make a program
- the stage: this is where the sprite moves
- the menu bar: you can choose commands from here.

Point to the different parts of the screen.

Draw the Scratch screen. Write the names of the different parts of the screen.

user **event**
command **forever loop**
random location

Talk about...

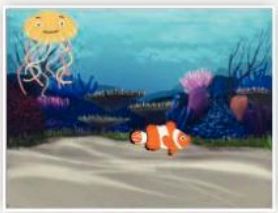
In this unit you will save your work. You must choose a file name. What makes a good file name? Talk to a classmate, then share your ideas with the rest of the class.



Did you know?

Scratch is used by children in more than 150 different countries. Scratch is available in different languages. You can make Scratch programs in more than 40 different languages.









4.1 Choose a sprite

In this lesson

You will learn:

- how to choose a sprite for your program.

Spiral back



Last year you used a simple program. It controlled a sprite. Now you will write a program to control a sprite.

In Unit 3 you played the Frog Hop Game. In this unit you will make a simple version of the game. You can choose any sprite you like. It does not have to be a frog.

You will make a Scratch program. Start on the Scratch website.

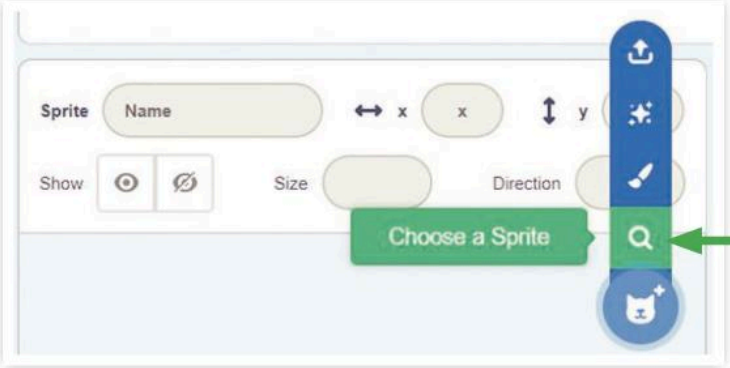
Choose a sprite

Look at the bottom right. This part of the screen shows the sprites.



Sprite 1 is a cat. Delete this sprite by clicking on the cross.

You will choose a new sprite.



Click here to see the other sprites.

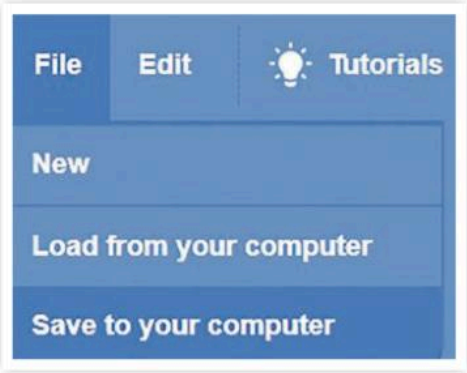
You will see pictures of sprites.



Click on the one you choose.

Save program file

Now you must save your work. You will save it on your computer. Open the File menu and choose 'Save to your computer'. Then type a name for your file.





Activity

Use the skills you learned to:

- pick a sprite
- save your file.



Draw the sprite you chose. Imagine a computer game with your sprite in it. Write down your idea.



Extra challenge

Add a backdrop to your program.

